

CHAPTER FIVE

THE CLIMATE OF ETHIOPIA AND THE HORN

Introduction

- Large country characterized by a wide variety of altitudinal ranges and diverse climatic conditions.
- Significant temporal and spatial variations in elements of weather and climate,
- Its climate is mainly controlled by the seasonal migration of **ITCZ** and **topography** of the country
- **Weather** - instantaneous or current state of the atmosphere composing temperature, atmospheric pressure, humidity, wind speed and direction, cloudiness and precipitation. Weather impacts takes place in the **troposphere**
- Hotness or coldness, rainy or cloudiness, sunniness, windiness or calmness on the daily base are expressions of weather.
- **Climate** refers the state of the atmosphere over long time periods, decades and more.

Elements and Controls of Weather and Climate

Elements	Controls
1. Temperature	1. Latitude/angle of the Sun
2. Precipitation and humidity	2. Land and water distribution
3. Winds and air pressure	3. Winds and air pressure
	4. Altitude and mountain barriers
	5. Ocean currents

- The climate of any place is determined by a combination of many interacting factors such as latitude, elevation, nearby water, ocean currents, topography, vegetation, and prevailing winds.
- Moreover, the global climate system and any changes that occur within it also influence local climate.
- **Latitude** is the distance of a location from the equator. The angle of the sun around equator is higher and as one moves further away from the equator towards the poles, less solar insolation is received during the year and the temperature become colder.

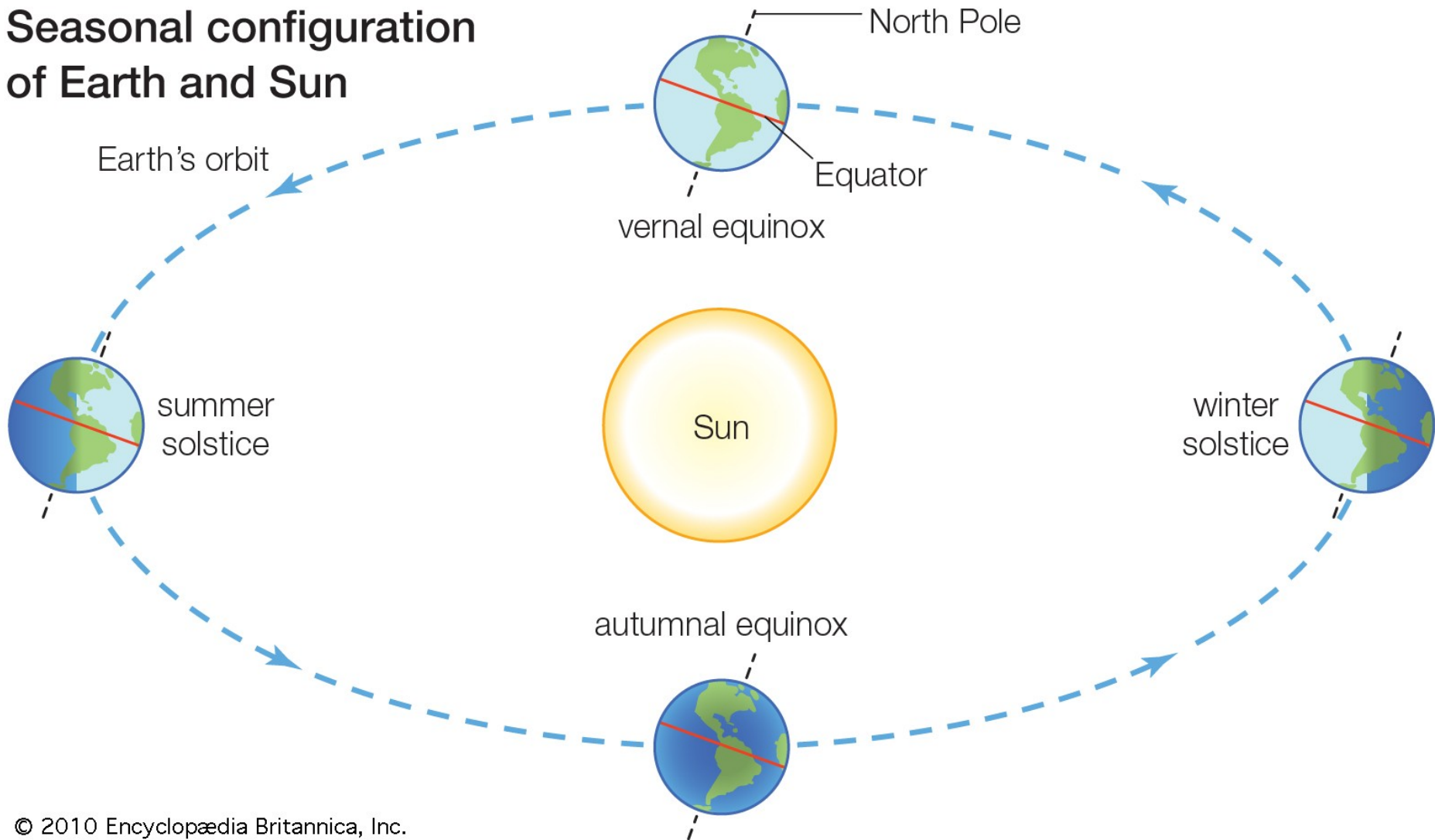
Latitudinal location of Ethiopia and the Horn resulted in;

- *high average temperatures,*
- *high daily and small annual ranges of temperature,*
- *no significant variation in length of day and night between summer and winter.*

Inclination of the Earth's Axis

- The earth's rotation axis makes an angle of about $66\frac{1}{2}^{\circ}$ with the plane of its orbit around the sun, or about $23\frac{1}{2}^{\circ}$ from the perpendicular to the ecliptic plane.
- Inclination of the axis determines the location of the Tropics of Cancer & Capricorn and the Arctic & Antarctic Circles.
- As the earth revolves around the sun, the inclination and constant position (parallelism) of the axis produces a change in the directness of the sun's rays; and differences in length of day and seasons.
- **Equinox** is the instant of time when the sun strikes the plane of the Earth's equator; and where the length of day and night are equal.

Seasonal configuration of Earth and Sun



Equinox appears twice a year.

The Vernal (spring) equinox: is the day when the point of verticality of sun's rays crosses the equator northwards.

- During this period, the length of day and night are equal.
- March 21 marks the offset of the vernal equinox.

The Autumn equinox: It appears when vertical sun's rays moves southwards on 23rd of September.

- ***Solstice*** is an event when the overhead sun is at its northern or southern extreme points resulting in unequal length of days and nights in the hemispheres.
- ***The summer Solstice:*** on June 21st, the northern hemisphere has maximum tilt towards the sun experiencing longest daylight of the year.
- The sun is at its highest position in the noonday sky, directly above $23 \frac{1}{2}$ in the Tropic of Cancer.
- ***The winter solstice:*** 22nd of December is the day when the maximum southward inclination is attained in the Southern Hemisphere.
- In the Northern Hemisphere, it occurs when the sun is directly over the Tropic of Capricorn, which is located at $23 \frac{1}{2}^{\circ}$ south of the equator.

Altitude

- Altitude refers to the height of location above the sea level.
- Under normal conditions there is a general decrease in temperature with increasing elevation in the troposphere.
- The average rate at which temperature changes per unit of altitudinal change is known as ***lapse rate***.
- The normal/***Environmental /atmospheric*** lapse rate is 6.5°C change per kilometer ascent/descent in altitude(if there is no temperature inversion).
- The lower layer of the atmosphere because:
 - atmospheric heat is received directly from the earth's surface and indirectly from the sun.
 - The lower layer has high density as it contains more water vapor and dust, which causes it to be a more efficient absorber of earth radiation than is the thinner, drier, cleaner air aloft.

Dry adiabatic lapse rate

- Refers to the temperature changes occurring in the rising or subsiding air mass;
- Occurs not as a result of additions of heat to, or withdrawals of heat from outside sources, but because of internal processes of expansion and contraction associated with a change in altitude.
- When air rises, it expands because there is less weight of air upon it.
- More precisely, if the upward movement of air does not produce condensation, then the energy expended by expansion will cause the temperature of the mass to fall at the constant *dry adiabatic lapse rate*. The rate of heating or cooling is about 10°C for every 1000 m of change in elevation.

Wet Adiabatic lapse rate

- The rate at which rising or sinking saturated air changes its temperature is less than the dry adiabatic rate due to the latent heat in the water vapor carried by the air.
- Prolonged cooling of air invariably produces condensation, thereby liberating latent heat.
- Rising and saturated or precipitating air cools at a slower rate than air that is unsaturated.
- This process is called wet adiabatic temperature change. The rate of cooling of wet air is approximately ***5°C per 1000 meters ascend***.

Spatiotemporal Distribution of Temperature

- Altitude (most important), latitude, humidity and winds have significant impacts on temperature conditions in Ethiopia.
- The location of Ethiopia at close proximity to equator, a zone of maximum insolation, resulted **for** every part of the country to experience overhead sun twice a year.
- Ethiopia, as it is a highland country & tropical temperature conditions have no full spatial coverage; they are limited to the lowlands in the peripheries.
- Mean annual temperature varies from over 30 °C in the tropical lowlands to less than 10°C at very high altitudes.
- The lowest & highest mean annual temperatures are recorded in Bale Mountains & the Afar Depression respectively.
- Lowlands of north-western, western and south-eastern Ethiopian experiences mean maximum temperatures of more than 30°C.

Temperature Versus Altitude

Altitude (meter)	Mean annual Temp (°C)	Description	Local Equivalent
3,300 and above	10 or less	Cool	Wurch
2,300 - 3,300	10 – 15	Cool Temperate	Dega
1,500 - 2,300	15 – 20	Temperate	Woina Dega
500 - 1,500	20 – 25	Warm Temperate	Kola
below 500	25 and above	Hot	Bereha

- The temporal distribution of Ethiopian temperature is characterized by extremes.
- Ethiopia's daily temperatures are more extreme than its annual averages.
- Monthly average temp. variation is minimal and the annual range of temperature is small.
- In the low seasonal temperature variations; months from March to June in Ethiopia have records of highest temperatures while low temperatures are recorded from November to February.
- Southern part of Ethiopia receives highest records of temperature in *autumn* and *spring* following the relative shift of the sun; whereas in the northern part of the country, summer season is characterized by higher temperature.

Spatiotemporal Distribution of Rainfall

- The position of Inter Tropical Convergence Zone (ITC), pressure cells, and Trade Winds determine the spatial and temporal rainfall variation.
- The inter-annual oscillation of the surface position of the ITCZ causes a variation in the Wind flow patterns over Ethiopia and the Horn.
- Following the position of the overhead sun, the ITCZ shifts north and south of the equator.

Seasonal or Temporal Variabilities

Summer (June, July, August)

- Most of Ethiopia except Afar and Southeast lowlands receive rainfall during summer
- High pressure cells develop on the Atlantic and Indian Oceans around the tropic of Capricorn
- During this season, Ethiopia and the Horn come under the influence of the **Equatorial Westerlies** (Guinea monsoon) and Easterlies that bring rain in this season.

Autumn (September, October and November)

- In autumn the ITCZ shifts towards the equator weakening the equatorial westerlies.
- During this season, the **south easterlies** from Indian ocean showers the lowlands in southeastern part of Ethiopia.

Winter (December, January and February)

- In winter, the overhead sun is far south of equator.
- During this season, **Northeasterly** winds originating from the landmass of Asia dominantly prevail Ethiopian landmass.
- The northeasterly winds crossing the Red Sea carry very little moisture and supplies rain only to the Afar lowlands and the Red Sea coastal areas.

Spring (March, April and May)

- In this season, the effect of the northeast trade wind is very much reduced; and the **southeasterlies** from the Indian Ocean provide rain to the highlands of Somalia, and to the central and southeastern lowlands and highlands of Ethiopia.

Rainfall Regions of Ethiopia

1. Summer rainfall region

- Comprises most parts of the country, except the southeastern and northeastern lowlands.
- The region experiences most of its rain during summer (kiremt), while some places also receive spring (Belg) rain.
- The High altitudes and the windward side experience rainfall of more than 1000mm.

2. *All year-round rainfall region*

- It has many rainy days than any part of the country.
- Found in the southwestern part of the country.
- Duration & amount of rainfall decreases as we move from southwest to north and eastwards; and summer months gain higher rainfall while the winter months receive lower amount.
- The average rainfall in the region varies from 1,400 to over 2,200 mm/year.

Autumn and Spring rainfall regions

- *South eastern lowlands* of Ethiopia receive rain during autumn and spring seasons under the influence of *southeasterly* winds.
- The average rainfall varies from less than 500 to 1,000 mm.

Winter rainfall region

- Red sea escarpments and some parts of the Afar region receive their main rain during the winter season from the northeasterly winds.

Agro-ecological Zones of Ethiopia

- Ethiopia possesses diverse agro-climatic zones that have traditionally been defined in terms of temperature.

1. The Wurch Zone (> 3,200 masl)

- Includes mountain systems of Ras Dashen, Guna, Megezez Shoa, Batu, Choke, Abune Yoseph etc.
- Receives mean annual temperature of less than 10°C.

2. Dega Zone (2,300 - 3,200masl)

- In Ethiopia, the Dega-zone is long inhabited and has dense human settlement due to reliable rainfall for agriculture and absence of vector-borne diseases such as malaria.

3. Weyna Dega Zone (1,500 – 2300masl)

- Has warmer temperature and moderate rainfall highly suitable for majority of crops grown in Ethiopia.
- It includes most of the agricultural land.

4. Kolla Zone (500 to 1500 masl)

- Climate of the hot lowlands In the geographic peripheries in south, southeast, west and northeastern part
- Average annual temperature ranges between 20°C and 27°C.
- Rainfall is low & erratic, but it could be as high as 1500 mm in the wet western lowlands of Gambella.

5. Bereha Zone

- Very hot, arid, windy climate of the desert lowlands.
- Largely confined to lowland areas with altitude of lower than 500 meters.
- *Its* average annual rainfall is less than 200 mm, and average annual temperature is over 27.5°C.
- Evapotranspiration is always in excess of rainfall.

Climate Change/Global Warming: Causes, Consequences and Response Mechanisms

- Climate change refers to a change in the state of the climate that can be identified (e.g. using statistical tests) by changes in the mean and/or the variability of its properties and that persists for an extended period, typically decades or longer.
- It refers to any change in climate over time, due to either natural variability or human activities.
- Ethiopian climate experiences extremes such as drought, flood etc.
- *A rise in average temperature of about 1.3°C has been observed between 1960 and 2006*
- Precipitation has remained fairly stable over the last 50 years when averaged over the country; though in some regions, total average rainfall is showing decline.
- Rainfall variability is increasing (and predictability is decreasing) in many parts of the country.

Causes of Climate Change

Natural Causes

- such as variations in the energy budget, the position of Earth relative to Sun, the position of continents relative to the equator,
- ***Earth orbital changes***: Changes in the tilt of the earth (23.5° to the perpendicular plane of its orbit) can lead to small but climatically important changes in the strength of the seasons. More tilt means warmer summers and colder winters.
- ***Energy Budget***: Although the Sun's energy output appears constant, small changes over an extended period of time can lead to climate changes (the Sun has been very gradually increasing its amount of radiation)
- ***Volcanic eruptions***: volcanic eruption releases large volumes of sulphur dioxide, carbon dioxide, water vapor, dust, and ash into the atmosphere that influence the climate

Anthropogenic Causes

- The warming of earth planet in the past 50 years is majorly driven by human activities generating greenhouse gases such as carbon dioxide, methane and nitrous oxide, *Chlorofluorocarbons (CFCs), water vapor.*
 - Industrial activities
 - Decomposition of wastes in landfills,
 - Agriculture, ruminant digestion and manure management,
 - Transportation emissions
 - Deforestation and land clearing

Consequences of Climate Change

- Climate change impacts are higher in low income countries that have limited capacity to cope with the changes.
- **Impacts on human health** (increased heat related mortality and morbidity, greater frequency of infectious disease, increased incidence of malaria...)
- **Impact on water resources:** melting of snow and glaciers that increases rise in sea level, increase drought and floods, distorted wind flow pattern, decreases water table, reduced amount of run-off into rivers, streams and lakes.
- **Impact on Agriculture:** effects on agricultural land, increases physiological stress and fodder quality and availability.
- ***Impact on Ecosystem:*** affects the success of species and community adaptation; even leading to species disappearance when unable to resist.

Climate Response Mechanisms

1. Mitigation and its Strategies

- Actions that are taken to reduce and control greenhouse gas emissions changing the climate; through reducing sources of these gases or enhancing the “sinks” that accumulate and store these gases(such as the oceans, forests and soil).
- Practice Energy efficiency
- Increase the use of renewable energy such as solar
- Efficient means of transport implementation: electric public transport, bicycle, shared cars etc.

Adaptation Strategies

- Implies adapting life in a changing climate through adjusting to actual or expected future climate.
- Meant to reduce our vulnerability to harmful effects of climate change such as extreme weather events or food insecurity.

Some of the major adaptation strategies include:

- Building flood defenses & water-permeable pavements to deal with floods and storm water ,
- Plan for heatwaves and higher temperatures,
- Improve water storage
- Landscape restoration and reforestation,
- Diverse cultivation to be prepared for natural catastrophes
- preventive and precautionary measures like evacuation plans

CHAPTER SIX

SOILS, NATURAL VEGETATION AND WILDLIFE RESOURCES OF ETHIOPIA AND THE HORN

Ethiopian Soils: Types, Degradation and Conservation

- Soil is a delicate but highly varied composition of mineral particles(45%), organic matter and living organisms in dynamic equilibrium(5%) as well as air (20-30%) and water (20-30%).
- The formation of a particular type of soil depends on parent material, climate, topography, living organism and time.
- Weathering disintegrates the inorganic substances (rocks) of soils.

Mechanical (physical) weathering

- Causes decrease in size without appreciably altering composition.
- Thermal expansion and contraction
- Abrasion (erosion by friction) due to water containing sediment or wind carrying debris
- Frost action
- Pressure release and unloading

Chemical weathering

- Involves the modification of the chemical and mineralogical composition of the weathered material.
 - hydrolysis,
 - oxidation,
 - carbonation,
 - and solution

Soil properties

Physical properties

- Include properties such as texture, structure, porosity etc. that affect air and water movement in the soil,

Chemical Properties

- Include soil properties like availability of minerals, electrical conductivity, soil pH, etc. that affect soil biological activity and indirectly the nutrient dynamics.

Major Soil Types in Ethiopia

- FAO has identified 18 soil associations in Ethiopia

1. Nitosols

- develop on gently sloping ground mainly from *volcanic rocks*
- associated with highlands with high rainfall and tend to be leached having reddish-brown color
- dominantly found in western highlands (*Wellega*), southwestern highlands (*Kaffa, Illuababora*), Southern highlands, Central highlands, and Eastern highlands.

2. Vertisols

- heavy clay highland soils; swell when wet, and crack when dry.
- Difficult to manage, but have high natural fertility.
- mostly develop on *volcanic plateau basalt & sedimentary rocks*.
- Commonly found in parts of *Northwestern, Central and Southeastern highlands* (n *Gojjam, Shewa, Arsi, Bale*).

Lithosols, Cambisols and Regosol

- These soils are mostly found in rugged topography and steep slopes.
- They are young, shallow and coarse textured and so have low water holding capacity.
- Found in areas of low rainfall.
- Have have limited agricultural use & in most cases, left under the natural plant cover and used for grazing.
- Found in rugged and steep slopes of Central Highlands, on the *Rift Valley Escarpments* and *highlands in of western Hararghe*, Danakil and eastern Ogaden.

Xerosols, Yermosols and Solonchaks

- These are soils of desert or dry steppe soils majorly available in arid and semiarid areas.
- Characterized by high salt content and low organic content, but are rich in phosphorus and potash and can be very fertile if irrigated.
- Xerosols are found in Ogaden and northeastern escarpments, whereas the Yermosols and Solonchaks cover the Ogaden and Afar plains.
- The Solonchaks are majorly located in salty plains of Afar.

Fluvisols

- Fluvisols develop on flat or nearly flat ground, on recent alluvial deposits.
- These soils are associated with *fluvial (river)*, *marine (sea)* and *lacustine (lake)* deposits.
- Found at lower regions of rivers like Omo, Awash, Abay and the plains of Akobo and Baro Rivers
- Fluvisols are much prized for intensive agriculture because:
 - they develop on flat ground, deposition sites,
 - they are associated with rivers and ground water, making them important for large-scale irrigation and
 - they are fertile and their fertility is always renewed as a result of deposition of new soil materials.

Luvisols

- Luvisols develop mainly in areas where pronounced wet and dry seasons occur in alternation.
- Where leaching is not very high, they are found in association with nitosols.
- Luvisols have good chemical nutrients and they are among the best agricultural soils in the tropics. So, they are intensively cultivated.
- Luvisols found on steep slopes (stony) and on flat areas (waterlogged) they are avoided and left for grazing.
- In Ethiopia, places with luvisols include *Lake Tana* area, parts of Northern, Central and Eastern Highlands and Southern lowlands.

Soil Degradation

- Refers to as a change in any or all of soil status resulting in a diminished capacity of the ecosystem to provide goods and services.
- Deterioration of the physical, chemical and biological properties of soil.

Soil degradation is a major concern for at least two reasons:

- First, it undermines the productive capacity of an ecosystem.
- Second, it affects global climate through alterations in water and energy balances and disruptions in cycles of carbon, nitrogen, sulfur, and other elements.

Types of soil degradation

- **Physical Degradation:** refers to the deterioration of the physical properties of soil through **Compaction**(densification of soil caused by the elimination or reduction of structural pores that makes soils susceptible to accelerated runoff and erosion) and **Soil erosion** (*detachment of individual soil particles, transportation and deposition by agents: water, air and glaciers*) . In Ethiopia, an estimated average of 42 tons per hectare of soils is eroded annually.
- **Biological Degradation:** Refers to reduction in soil organic matter content, decline in biomass carbon, and decrease in activity and diversity of soil fauna. It can be caused by indiscriminate and excessive use of chemicals and soil pollutants.
- **Chemical Degradation** entails nutrient depletion as a result of excessive leaching of cat-ions, and changes in soil pH

Causes of soil degradation

- Natural and human-induced causes of soil degradation:
 - Topographic and climatic factors such as steep slopes,
 - Frequent floods and tornadoes,
 - Storms and high-velocity wind,
 - High-intensity rains and drought in dry regions
 - Deforestation and overexploitation of vegetation,
 - Overgrazing,
 - Indiscriminate use of agrochemicals
 - Lack of soil conservation practices,

Soil Erosion Control Measures

- Since erosion is a natural process, it cannot be prevented. But it can be reduced.

1. **Biological Control measures** reduce the velocity of surface runoff, increase infiltration and reduce runoff

- ✓ vegetative strips,
- ✓ plantation, and reforestation. and etc.

2. **Physical control measures**

- ✓ terracing,
- ✓ trenches,
- ✓ contour ploughing,
- ✓ soil bunds
- ✓ Wind breaks or shelter belts
- ✓ check dams,

Natural Vegetation of Ethiopia

- *Natural vegetation* refers to a plant cover that develops with little or no human interference.
- Its distribution on the surface of the earth is uneven majorly controlled by factors such as *climate, soil types, drainage*, etc.
- Uses of natural vegetation:
 - ✓ shelter, food,
 - ✓ source of fuel,
 - ✓ pasture and grazing,
 - ✓ raw material for industries,
 - ✓ source of timber and non-timber products.
 - ✓ moderating effect on local climate,
 - ✓ home of wild life,
 - ✓ medicinal values,
 - ✓ minimizing soil erosion

- The characteristics of Ethiopia's natural vegetation are largely determined by **elevation** (and temperature) and rainfall.
- Lowlands due to their low rainfall and high temperature have harsh environment and are characterized by *xeromorphic* plants (plants which are adapted to drought and high temperatures).
- Highlands (up to about 3000 m) are cooler than the lowlands and where there is adequate moisture a variety of forests can be seen.
- In highlands of greater than 3000 masl, tree growth is limited by the low temperature
- Ethiopia possesses an estimated number of 6000 species of higher plants of which 10% are endemic.

Major Natural Vegetation Types of Ethiopia

Vegetation belts of Ethiopia base on altitude:

- ***Afro-alpine and sub-afro alpine Region***
- ***Forest Region***
- ***Woodland Savannah Region***
- ***Steppe Region***
- ***Semi-desert Region***

Afro-alpine and Sub-afro alpine Region

- Ethiopia has the largest extent of Afro-alpine and sub afro-alpine (high mountain vegetation) habitats in Africa.
- Found on mountains where elevation is greater than 3,200 masl.

The Afro-alpine region:

- ❖ found at very high altitudes (4,000 – 4,620 m).
- ❖ annual precipitation ranges between 800 and 1,500 mm, mostly in the form of sleet or snow.
- ❖ Temperature records of 0°C and below are widely experienced
- ❖ Soils are mostly shallow and eroded.
- ❖ Vegetation in the Afro-alpine region consists of tussock grasslands, scrub, scattered mosses and lichens
- ❖ *Lobelia rhynchopetalum* (gibbera) is the dominant species in the Afro-alpine.

Sub-afro-alpine region :

- ❖ found at a lower elevation, roughly between 3,200 and 4,000 meters.
- ❖ plants in this region are adapted to somewhat less extreme environment than the **Afro-alpine**.
- ❖ dominated by woodland, often degraded to scrub stages and also wet grasslands.
- ❖ *Erica arborea* (Asta) is the dominant species in the Sub-afro alpine region.

Forest Region

- Forest is a complex ecosystem consisting predominantly of trees that shield earth and support numerous life forms.
- The type of forests in any geographical region is determined by climate, soil types, topography and elevation .
- In Ethiopia, forests are found at different elevations (450 - 3,500m in humid parts & 2,300 - 3,300 m in arid parts); and are characterized by variation in mean annual rainfall that range between 200 and 2,200mm.

Classification of forests:

- ❖ **Highland forests:** Hagenia Abyssinia (*Kosso*), Juniper procera (*tid*), Arundinaria Alpina (*kerkha*), Podocarpus falcatus (*zigba*), Aningeria adolfi-friedericicii (*keraro*) and Olea africana (*Weyra*) forests;
- ❖ **Lowland forests:** Baphia, *Gallery (Riverine) Forests* that stretch along the banks of the lower courses of rivers, dominant species include Ficus sur (*sholla*) and different kinds

Woodland Savannah Region

- found in areas of wide altitudinal ranges (250 to 2,300 m).
- The plants in the woodland savannah are known for their **xeromorphic** characteristics like shading of leaves during the dry season.

Divisions:

- ***Juniper procera* (tid)** is dominant species for both the Junipers Forests and Junipers Woodlands. The difference is in height: 3 - 45 meters tall in the forests and 10 -15 meters in the woodlands.
- ***Acacia woodlands*** are dominated by both trees and shrubs, which belong to the same genus 'Acacia'. E.g. **Acacia etbaica(grar),Acacia mellifera (Konter)**.
- ***Mixed deciduous woodlands***: consists of trees that mainly shed their leaves during the dry season.

Steppe and Semi Desert Regions

- These are regions in the arid and semiarid parts of the country where temperature is very high and rainfall very low.
- soils in both regions are generally alkaline and saline.
- In these regions xerophytic plants (drought-resisting) such as short shrubs, scattered tufts of grass species and a variety of acacias are dominant.
- Trees are normally restricted to fringes along watercourses.

The steppe is found at elevations of *100 to 1,400 msal*

- The steppe gets a mean annual rainfall of *100 to 550 mm*
- Growing period lasts up to 2 months for the steppe

Semi-deserts are found at *125 masl to 600 masl*.

- get a mean annual rainfall of *50 to 300 mm* .
- Growing period is a maximum of one month .

Natural vegetation Degradation

- Over the past century, a rapid growth of the population has led to overexploitation of the forest land; reducing the coverage from about 40% to less than 10%.
- Major causes of forest degradation:
 - Clearing of forests for cultivation
 - clearing for construction
 - Timber exploitation
 - Charcoal burning and cutting for fuel
 - Extensions of coffee and tea production areas
 - Overgrazing
 - Expansion of settlements.

Natural Vegetation Conservation

- There is an urgent need of conservation of the ever-degrading biodiversity.
- There are three main approaches of biodiversity conservation:
- ***Protection***: through designation and management of some form of protected area such as sanctuaries, national parks, and community conservation areas.
- ***Sustainable forest management***: involving sustainable harvesting of forest products to provide a source of financial income
- ***Restoration or rehabilitation***: is the process of assisting the recovery of a forest ecosystem that has been degraded, damaged, or destroyed.

Wild Life/wild animals in Ethiopia

- One of the few countries that possess diverse fauna with a high level of endemism because of wide range of ecosystems.

Wild animals are useful for:

- scientific and educational researches (valuable information for medical purposes and environmental studies)
- physical and mental recreation (aesthetic value)
- promotion of tourism (economic value)
- its potential for domestication
- maintaining ecological balance

Ethiopia has:

- about 860 avian species (16 endemic species and two endemic genera),
- 279 species of mammals (31 endemic species and six endemic genera), i.e. Walia Ibex (*Capra walie*), Gelada Baboon (*Theropithecus gelada*), Mountain Nyala (*Tragelaphus buxtoni*), Ethiopian Wolf (*Canis simensis*), Starck's Hare (*Lepus starcki*)
- 201 species of reptiles (14 endemic species),
- 23 species of amphibians (23 endemic species),
- and 150 freshwater fish (6 endemic species).

- The main wild life concentrations in Ethiopia occur in the southern and western parts.

Classifications of wild animals in Ethiopia:

1. **Common wild animals:** those found in many parts of the country like *hyenas, jackals*)
2. **Game (lowland) animals:** include many *herbivores like giraffes, wild asses, zebras etc. and carnivores like lions, leopards, and cheetahs*
3. **Tree animals or arboreals :** *monkeys, baboons*
4. A variety of *birds* in the Rift Valley lakes
5. **Rare animals** (*gelada baboon and Semien fox*) scattered in highlands; *walia- ibex* in the Semien Massifs, *Nyala* in the Arsi Bale massifs).

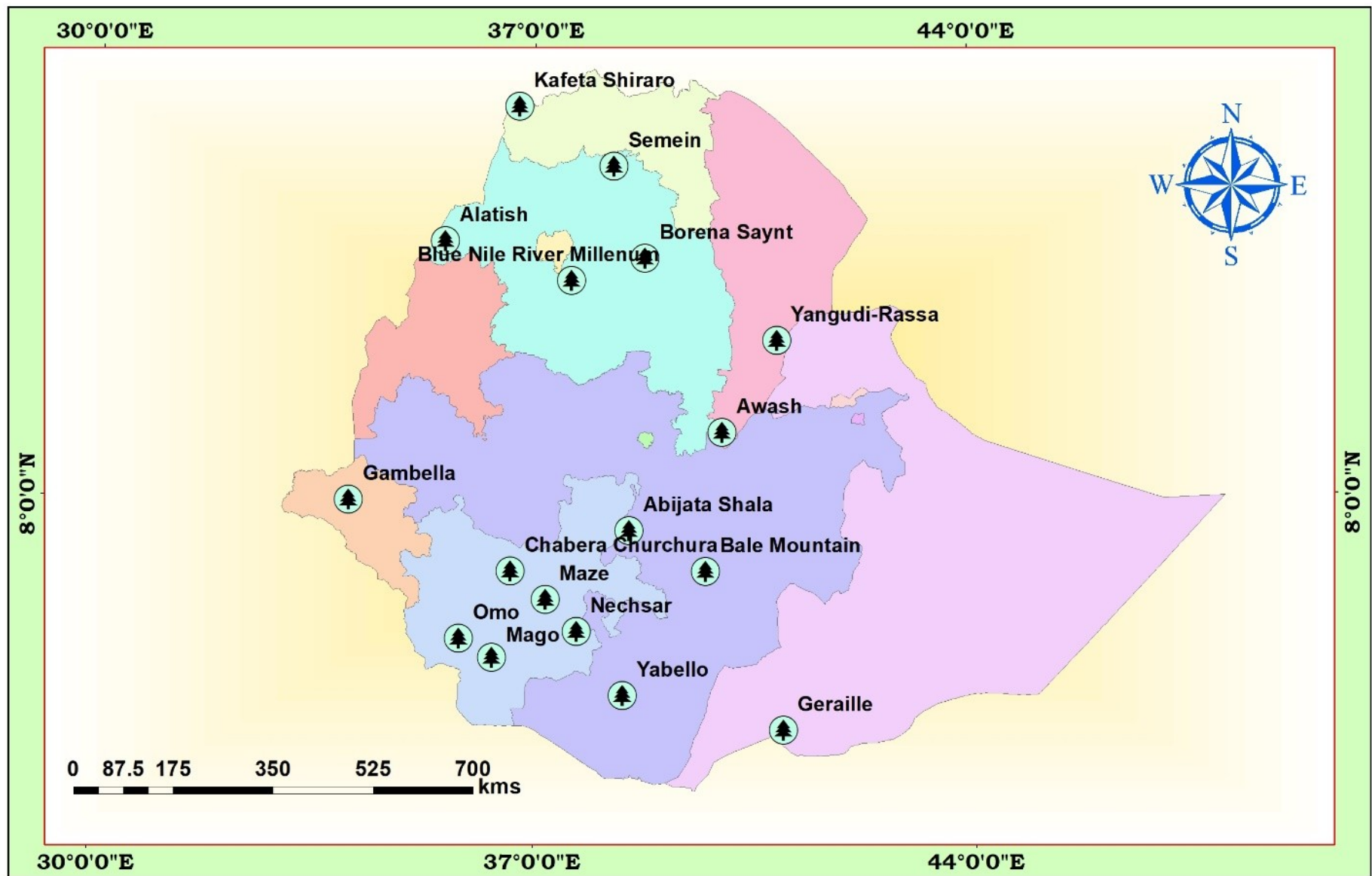
Wildlife Conservation

- Conservation of Ethiopia's biodiversity and ecosystems is vital to ensure sustainable development, mitigate and adapt to the effects of climate change and prevent the collapse of life-supporting ecosystem services. As such we have:

21 major national parks ,	2 wildlife rescue centres
2 major wildlife sanctuaries	22 controlled hunting areas
3 wildlife reserves,	2 botanical gardens, and 3 biosphere reserves
6 community conservation areas	

Uniqueness of national parks in their wild animals:

- ***Abiyatta-Shalla lakes National Park*** : predominantly bird sanctuary, important bird species include flamingos & pelicans.
- ***Omo, Mago, and Gambela National Parks*** have hippopotamus and crocodiles in rivers and lakes.
- ***Semien and Bale Mountains National Parks*** have rare animals like *Walia ibex*, *Semien fox*, *gelada baboon* and *Nyala*.



Spatial distribution of National Parks

Challenges of wildlife conservation in Ethiopia

- Limited awareness on the importance of wild life
- Expansion of human settlement in protected areas.
- Conflict over resource
- Overgrazing
- Illegal wildlife trade
- Excessive hunting
- Tourism and recreational pressure
- Mining and construction material extraction
- Forest fire

CHAPTER SEVEN

POPULATION OF ETHIOPIA

Population:

- In biology population refers to the number of organisms of the same species
- In statistics, a population is the entire pool from which a statistical sample is drawn.
- For this course population could mean the total number of persons/people inhabiting a country/city/or any geographic area.
- Human population can be considered as a point of reference from which all the other elements are observed, and from which they all, derive their significance and meaning.

- **Many disciplines** like Geography, Demography, Economics, Epidemiology, Sociology ... study human population. Their differences lie in the methodologies they employ and the aspects they emphasize
- The study of population in Geography aims at showing and explaining regional/spatial differences on population **distribution** and densities, population **numbers**, human-environment **interactions**, population **dynamics** (fertility, mortality and migration), as well as population **characteristics** and **qualities** (age, sex education and health composition etc.).

Population Data: Uses and Sources

- Regular and reliable population data are vital for effective socioeconomic development planning and administration.
- Demographic data are crucial to administrators, businessmen, researchers, academicians and planners
- Demography influences production, distribution, consumption, defense and administrative services at any unit considered.
- Population data are needed to plan for the provision of infrastructures and services such as schools, hospitals, roads, water and sewerage facilities, housing, establishing voting district boundaries, estimating future tax revenue and designing public programs.

Conventional sources of population data

A. Census

- The total process of collecting, compiling and publishing demographic, economic and social data pertaining at a specified time (s) to all persons in a defined territory.

Characteristics of a census:

- **Universality:** all persons in a given area during the count,
- **Periodicity:** census undertaking at regular time intervals with reference to a defined point of time usually 10 years and 5 years,
- **Simultaneity:** undertaking census in a very limited time duration called the census day/night,
- **Government sponsorship** and publication

Procedures for collecting census data: de jure and de facto

- **De jure approach:** counting people according to their usual place of residence (where he/she lives most of the time).
- **De facto approach:** each individual is recorded at the place where he/she was found at the time of the census.

B. Sample Survey

- This is a method in which a defined population/sample/ is selected with the view that information acquired would represent the entire population.

Advantages of sample survey over census:

- costs can be greatly reduced;
- simple to administer and taken much faster
- Used to obtain detailed information to supplement census data.

However, sample surveys have the inherent weaknesses related to sampling errors and inadequate coverage .

- Data from most censuses and sample surveys include geographic location, age, sex, marital status, citizenship, and place of birth, relationship to the head of household, religion, educational characteristics, occupation, fertility, income, language, ethnic characteristics, disabilities and migration.

C.

Vital Registration

- Vital registration is a system of continuous, permanent, compulsory and legal recording of the occurrence and the characteristics of vital events like births, deaths, marriages, divorces, and adoptions.
- Population data errors: **coverage and content**
- **Sources of Population data errors:**
 - **poor and inadequately financed methods of collection;**
 - **poorly trained enumerator;**
 - **suspicion and ignorance of censuses and false statements specially of age and income;**
 - **constant changes in administrations;**
 - **omission of more inaccessible areas;**
 - **as well as wide difference in connotation of terms like language, ethnicity, and occupation.**
 - As such, the errors need to be **detected** and all the necessary adjustments made to enhance their usefulness.

Population Dynamics: Fertility, Mortality and Migration

- The population of any particular region (country) grows/declines as a result of the combined effect of the three demographic variables: fertility, mortality and migration.
- Ethiopia is endowed with a large and fast-growing population ranking 2nd in Africa after Nigeria.
- According to the first ever census return of 1984, the population of Ethiopia was 42.2 million. Based on projected data from CSA, Ethiopia has an estimated 2019 population of 112 million, which ranks 12th in the world, and the current growth rate is about 2.6%.
- The largest part of the population (80%) is rural based.
- Ethiopia has a youthful population where about 65 percent of the population is below 24 years of age.
- The population is also unevenly distributed.

Demographic Measurements

- In Ethiopia, fertility and mortality are the two principal determinants of population growth.

1. Crude Birth Rate refers to the number of live births per 1000 population

$$CBR = \frac{\text{Total annual live birth}}{\text{Total midyear population}} \times 1000$$

2. General Fertility Rate refers to the total number of live births per women of reproductive age

$$GFR = \frac{\text{Total annual live birth}}{\text{Total women 15 - 49}} \times 1000$$

- **GFR** is a relatively specific measure of fertility as it specifically relates births to women in the reproductive age.

3. Total Fertility Rate (T.F.R.) refers to the average number of children that a woman would have at the end of her reproductive period if the current age specific fertility rate remains unchanged.

$$TFR = 5x \sum_{n=1}^7 \frac{Bi}{Wi}$$

4. Crude Death Rate refers to the number of deaths per one thousand population in a year

$$CDR = \frac{\text{Total annual Death}}{\text{Total midyear population}} \times 1000$$

5. Infant Mortality rate refers to the total number of deaths of infants per one thousand live Birth

$$IMR = \frac{\text{Total annual infant death}}{\text{Total live birth}} \times 1000$$

- Infant mortality draws special attention because of its large size and the heavy impact it has on the crude death rate. The 2016 demographic and health survey result showed that infant mortality in Ethiopia has decreased from 97 deaths per 1,000 live births in 2000 to 48 in 2016.

6. Maternal Mortality Rate: refers to death of mothers in connection from pregnancy and birth complications per hundred thousand live birth.

$$\text{MMR} = \frac{\text{Total annual maternal death due to birth complications}}{\text{Total live birth}} \times 100,000$$

7. Life Expectancy at birth: refers to the average number of years that a newly born baby is expected to live. It is used as a summary measure of the mortality experience of the whole population.

8. Natural Rate of Increase: is the difference between crude birth rate and crude death rate expressed in percentage.

- $\text{NRI} = (\text{C.B.R.} - \text{C.D.R.})$

Levels and trends in Fertility and Mortality rates in Ethiopia

- Birth and death rates show significant spatiotemporal variation.
- Clear differences in birth and death rates are emerging between rural and urban areas of Ethiopia (Eg. Rural TFR is 5.2 compared to 2.3 for urban).
- There is regional variation: TFR by region in 2016 was the lowest in Addis Ababa (1.8) followed by Dire Dawa (3.1), Gambella (3.5), and Amhara (3.7); while Somali (7.2), Afar (5.5), Oromia (5.4), and Tigray (4.7) had higher rates.
- Recently, fertility is showing a declining trend. Total fertility rate (TFR) declined from 7.52 in 1984 to 6.74 in 1994, and currently, at 4.6.
-

- Mortality rates are also showing a **declining trend** (For example before 2000, all regional states had IMR of more than 100, but by 2011 infant mortality in all regions was lower than 100, except for Benishangul Gumuz.
- Mortality rates also show considerable **variation by regions**. (In 2016, IMR at the country level was 54 where it was 48 in urban areas and 62 in rural Ethiopia)
- Lower than national average **infant mortality** rate was recorded in Addis Ababa followed by Somali and Gambella; while higher IMR was recorded in Benishangul, SNNPR and Tigray.
- **Life expectancy at birth** in Ethiopia increased from about 36.7 years in the 1960s to 62.6 years in 2016.
- Life expectancy: (Monaco=89.4; Ethiopia=62.6 to Chad=50.6)
- IMR: (Afghanistan=110; Ethiopia=49.6 to Monaco=1.8)
- CDR: (Lesotho=15.0; Ethiopia=7; to Qatar=1.5)
- CBR: (Angola/Niger= 44.2; Ethiopia= 36.5; to Monaco=6.6)

- As opposed to declining death rates, birth rates have remained high due to:
 - Little family planning practices and lack of population education;
 - Lower status of women
 - Early marriage, particularly of females;
 - Parents consideration of children as assets, though little obliged for their education, health;
 - The relatively high infant and child mortality rates, that trigger couples to have more births to compensate for the losses
 - Perhaps religious influences.

- Our population is growing rapidly. Some of the consequences of this rapid population growth include:
 - ❑ low per capita GNP
 - ❑ increased unemployment and under –employment
 - ❑ mounting social ills such as destitution, begging, theft, prostitution
 - ❑ continuous inflation that erodes purchasing power of the currency
 - ❑ shortage of cultivated land and food shortages
 - ❑ overcrowding of infrastructural and social facilities; housing problems and increase in urban slums and squatter settlements
 - ❑ Environmental problems such as deforestation, soil erosion, loss of biodiversity and pollution.

Migration in Ethiopia

- Migration is an old and inevitable phenomenon, although accelerated these days as a result of improvements in communication and transportation.
- Geographic mobility involving a permanent or semi-permanent change of residence between clearly defined geographic units.
- Implications of migration:
 - ✓ Migration yields an increased level of urbanization;
 - ✓ It enhances rural-urban linkages
 - ✓ It influences spatial population distribution
 - ✓ influences fertility, mortality & age and sex composition.
 - ✓ It is a means of achieving economic efficiency.
 - ✓ cause and consequence of inequality & unequal development
 - ✓ cause & consequence of diversity; mechanism of spreading cultures
 - ✓ strengthens of a sense of nationhood and national unity
 - ✓ creates a creative and open society to new ideas

Internal Migration in Ethiopia

- Ethiopia is an ancient original abode of human migration
- Population movement accelerated in the early twenty century with the **rise in urban centers** as well as the **Italian occupation**.
- **Rural out migration during the Derg Regime was low for:**
 - The 1976/77 '**land to the tiller**' granted land to the rural landless farmers, that reduced their motivation for migration.
 - A person should be either a member of an urban dwellers association or rural peasant associations
 - The 1975 urban land nationalization dispossessed landlords' rights to own more than one house that further led to a chronic shortage of urban **housing** .
 - The high level of urban **unemployment** and underemployment was a disincentive for potential migrants.
 - The Derge was also taking away whoever is scrounging around in the city as soldiers to the **warfront** .

- Despite the high unemployment and ethnic politics that discourage inter-regional migration; rural outmigration is still high due to :
 - Increasing household size, diminution of landholdings and lack of farmland;
 - Drought, rainfall unreliability, & land degradation that diminished carrying capacity of the land
 - Indebtedness and the need to payback
 - Lack of non/off-farm income
 - Dissatisfaction with village life & glamour of towns
 - Improvements in rural education, transportation and communication
 - Marriage related and family disintegration
 - Peer influence
 - Presence of other family members in a destination
- **Changing community attitudes to migration**
- Internal migration in Ethiopia is among the highest in Africa; about 16.6% is labeled as migrant population.

International migration

- International migration in Ethiopia accelerated after the 1974 revolution where many refugees were attempting to escape political conflict, persecution and famine.
- Today, Ethiopia could be considered as one of the countries that has a large number of emigrants overseas.
- Ethiopia's diaspora, estimated to be about four million, is also considered one of the largest of all African countries.
- Large numbers of Ethiopian migrants are found in the Middle East, USA, Canada, Europe and African countries such as Sudan, Kenya, South Africa and Botswana.

The causes of cross-border migration include:

- Unemployment, negative attitudes attached with low paying and informal job, and poor work ethics amongst the youth.
- Rural underemployment and lack of resources
- Unfavorable political context and insecurity,
- Location in the fragile region of the Horn of Africa and its long boundary (5,328 km) which makes border management difficult
- Existence of large number of local brokers with networks extending to countries of destination;
- Misinformation & false promises by brokers/traffickers; success stories of pioneering migrants; family & peer pressure
- Emergence of 'culture of migration' and migration networks
- **Demand-side factors** of migration (shortage of labour in low-paying, informal, and perilous jobs, such as domestic work, construction, agriculture in destination countries)

- Ethiopia is a country of origin, transit and destination for international migration.

Hub on three land routes:

- one leads from the Horn of Africa via Sudan, Chad, Egypt and Libya to the Mediterranean Sea towards Europe;
- the second through Somalia and Djibouti to Yemen across the Gulf of Aden and Red Sea towards the Middle East. 60 to 70% of Ethiopians migrating to the Middle East are irregular migrants.
- The third migration route is the Southern irregular route that is an overland route Kenya Tanzania towards to South Africa.

Bole International Airport is also reported to be a hub of transit on the air route leading to Europe, and the Middle East.

Emigration source areas of Ethiopia Organization/ILO/ in 2016

1. **Dessie** (North and South Wollo) area: it includes Kemise, Bati, Kalu (Kombolcha), Dessie and its surroundings, Tehuledere (Haiq, Girana, Bistima, Bakaksa, Worebabo), Mersa, and Woldia.
 2. **Shashemene** (Western Arsi and Bale) area: it include Shashemene-Zuria, Kofele, Kore, and Assassa.
 3. **Jimma** (Western Ethiopia) area: includes Kaffa, Wolega and Iluababora, and more specifically Mana, Kerisa, Dedo, Agaro, Setema, Sigmo, and Gomma areas.
 4. **Mekelle/Tigray** area: specific localities include Alamata, Kobo, Raya, Erob, Edagahamus, Gulomehadi, Etsebi, and Weneberta.
- Other prominent emigration source areas include (Assela-Zuria, Adama-Zuria, Ambo, Fitcha, Chanco, and Western Hararghe (Hirna, Gelemso); Shewa Robit, Debre Birhan, and Debre Tabor).

Age and Sex Structure of Ethiopian Population

- What one is, thinks, does, and needs is closely related to the number of years since he/she was born.
- Production, military services, political propensities, and mobility are influenced by age composition
- Age Structure refers to the distribution of population by age groups.
- The most used age groups are five-year age groups (0-4, 5-9, 10-14, ..., 60-64, 65 and above) and broad age groups (0-14, 15-64, 65 and above).
- Age groups 0-14, 15-64 and 65 and above are known as young age, working age and old age, respectively.
- It is generally accepted that people in the young and old ages are dependent on the working age population.

- Age structure is influenced by fertility, mortality and migrations.
- ✓ A population with a high birth rate will have a large proportion of young people
- ✓ A population with a high death rate will have lower proportion of older people.
- ✓ An area that experiences a great deal of out migration usually has fewer young people (20-40 years)
- Age **dependency ratio** measures the role of age composition on the productive activity of a population by comparing the proportion of the population in the non-productive ages with those in the working-ages.

$$A.D.R = \frac{P_{0-14}}{P_{15-64}} \times 100$$

➤ Youth Dependency Ratio: $\frac{P_{0-14}}{P_{15-64}} \times 100$

P15 -64

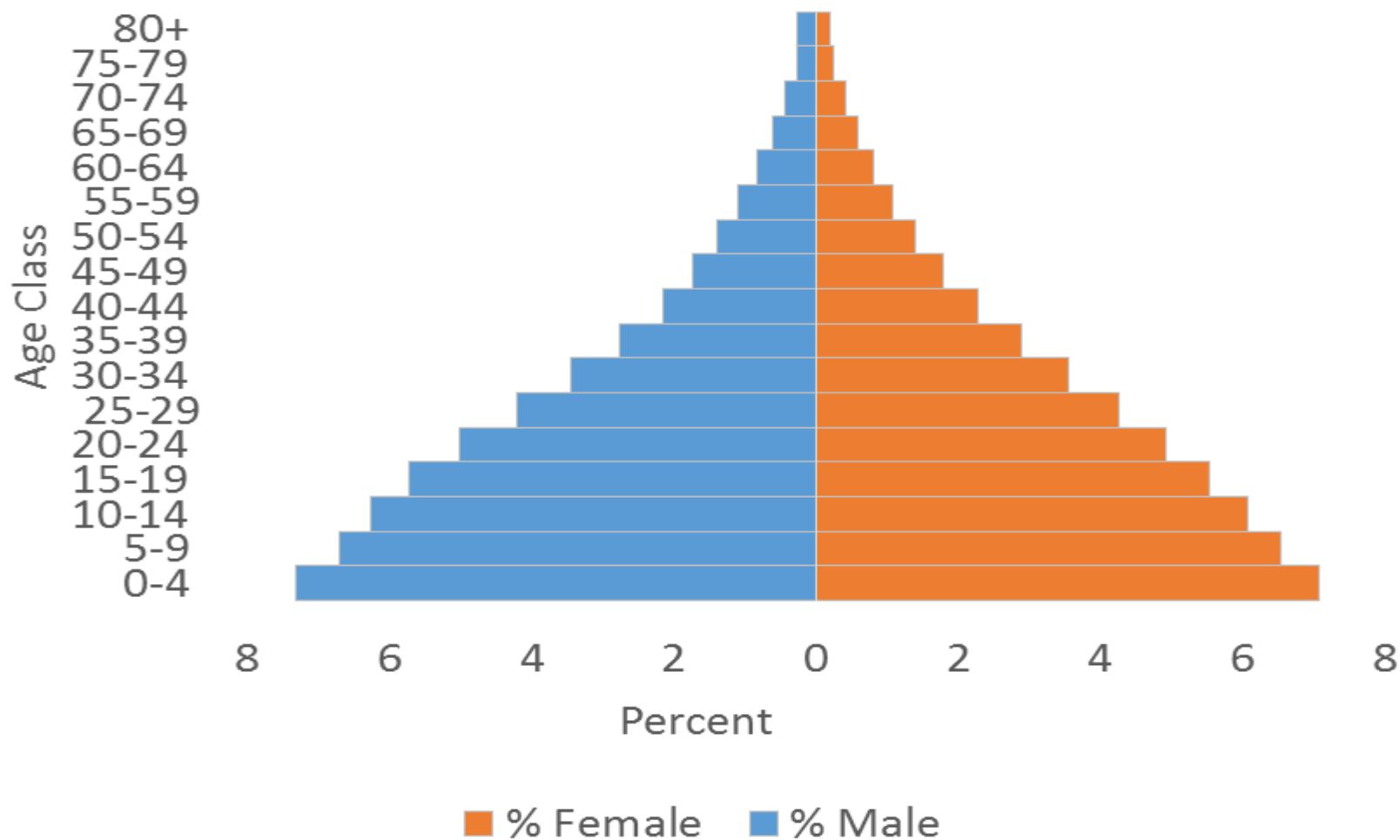
- Ethiopia has a youthful population and heavy youth dependency. Youth dependency ratio for Ethiopia's population in 2007 was about 93

Implications of heavy youth dependency:

- Imposition of heavy burden on the working population
- allocation of most of the household budget to food and other household needs with little/nothing left for saving; which then affects investment
- diversion of limited resources on social services - building of schools and hospitals and purchase of medicines, etc. which could have been geared to directly productive investment; and
- creation of a society with booming babies that require an expansion of employment opportunities by the time they are of age
- further promotion of high-level fertility by increased number of women entering the reproductive age (ages 15-49) annually.

Population Pyramid

- Population pyramid (age pyramid) is a graphic representation of the distribution of the population by age and sex.
- In the age pyramid, the vertical axis is graduated in years (usually five-year age groups) from zero upwards and the horizontal axis shows the percentages of males and females within these groups.
- Percentages are calculated for both sexes separately or of the total population.
- **A Population pyramid for a country:**
 - ✓ represents its demographic history over the past two or three generations
 - ✓ demonstrates the sex differential in age structure
 - ✓ shows which age groups have large number of people
 - ✓ depicts the fertility and mortality experience of the population
 - ✓ indicates the future growth potential of the population.
 - ✓ implies pattern of consumption and governmental obligations and expenditures.
 - ✓ shows the short and long terms effects of wars and migrations



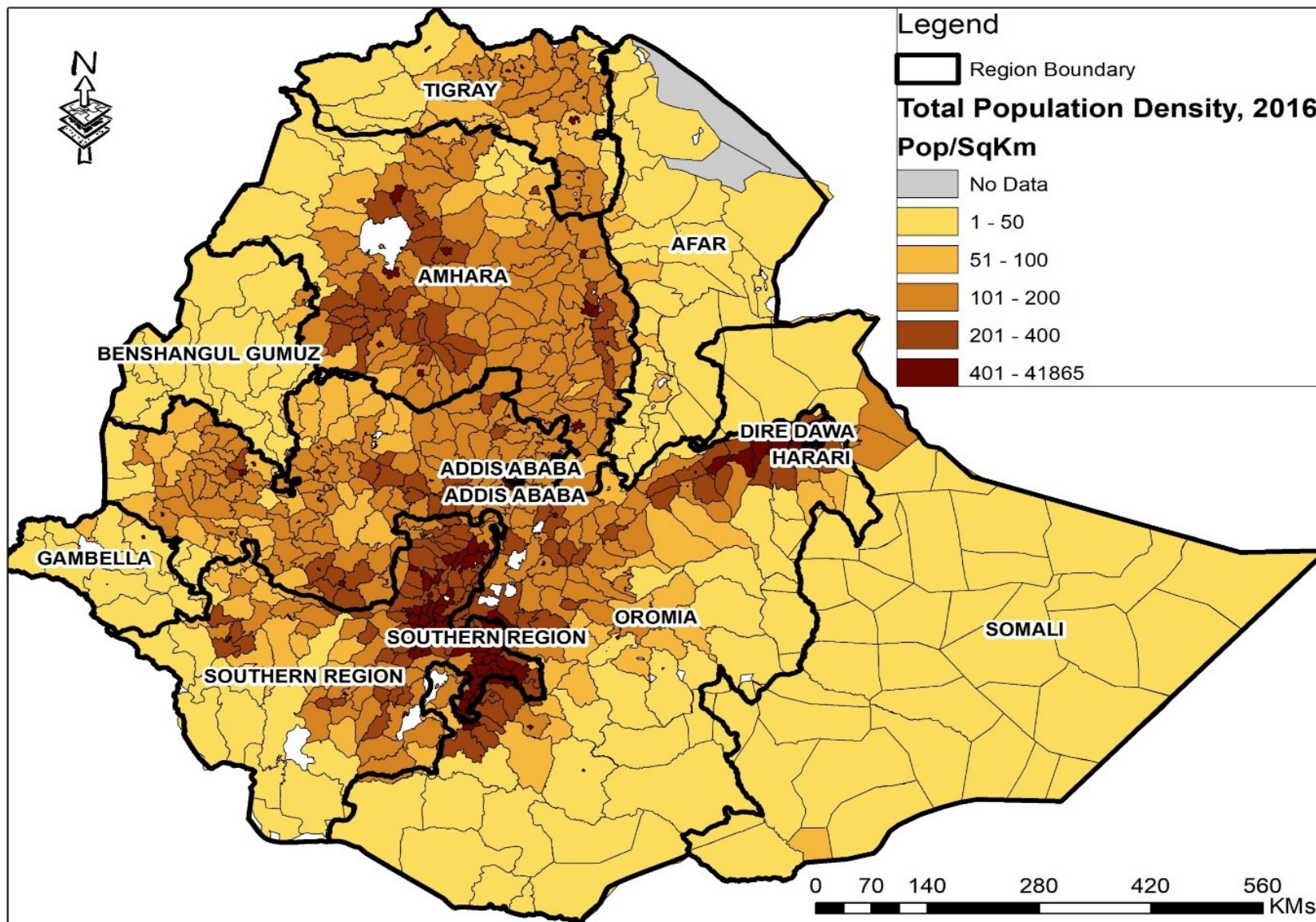
Population pyramid of Ethiopia, 2017

Sex Structure

- Sex structure refers to the ratio of male population to female population at different age groups.
- It is usually expressed as:
$$\text{Sex ratio} = \frac{\text{Male}}{\text{Female}} \times 100$$
- In 2015, sex ratio for Ethiopia was 99.96.
- Sex ratios are generally lower for urban areas, and higher for rural areas primarily due to larger female in-migration to urban areas.
- Sex composition of the population also shows some variation by region. In Afar, Somali and Gambella, the number of males exceeds that of females, while in Addis Ababa the number of females is considerably higher than the number of males.
- Sex ratios also vary with age. Sex ratios are high in young age groups and low in adult and old age groups.

Population Distribution in Ethiopia

- Population distribution refers to the arrangement of people over space that is provided for them to settle and make a living through exploiting resources.
- The distribution of population in Ethiopia is very **uneven** as a result of physical and human factors operating together.
- **Factors of population distribution:**
 - climate,
 - soil,
 - vegetation,
 - drainage and slope
 - historical pattern of population movement,
 - type of economic activity,
 - urbanization, industrialized
 - demographic variables



Population Distribution Map of Ethiopia

Measures of Population Distribution: Population Density

- Population density refers to the number of people per unit area.
- 1 **Crude density:** found by dividing total population to total area.
- It does not show variations in population distribution within a given area.
- It amounted to more than 100 currently.
- It shows considerable regional **variations**. Excluding the urban based administrative regions, Southern Nations, Nationalities and Peoples (SNNP) region is the administrative area with the largest population density (173 people/km²) followed by Amhara region. Gambella (13 people/km²), Somali, Afar and Benishangul-Gumuz are regions with low densities of population.
- The variation in crude density could be more pronounced if we consider subordinate administrative units in the hierarchy

- **Physiological Density:** is a ratio between total population and arable part of a country. Ethiopia's physiological density (for 1998) is 61.8 people/km².
- Compared to Physiological densities of countries like Japan (1,732 people/km²), Egypt (1,575 people/km²) and Netherlands (1,220 people/km²), Ethiopia's physiological density is very low.
- **Agricultural Density:** takes only agricultural population as a numerator and cultivated land as a denominator.
- It is also called rural density since in most developing countries there is not a significant difference between rural and agricultural population.
- Rural population per square kilometer of cropland is the highest for Somali, Afar and Gambella. SNNP region has also a high ratio compared to the national average owing to the high population relative to their agricultural land. The smallest agricultural density lies in Benishangul (3.8), Amahra (5), Oromia (5.6) and

Socio-cultural Aspects of Ethiopian Population: Education, Health and Languages

- Despite recognizable improvements, the health and education condition in Ethiopia is at a staggering situation
- **Education**: high illiteracy levels, low but growing number of schools, high dropout, overburdened facilities, lower female participation in some regions... despite recognizable improvements
- **Health**: poor health status of the population:
 - Lack of access to clean water
 - Lack of adequate nutrition
 - Disease related to beliefs, behaviors and traditional practices such as circumcision, early marriage, and low value of girls and children
 - Lack of health services: health service coverage is less than 50% of the population that tend to be urban biased.

Languages Families and Languages of Ethiopia

- Ethiopia is a country where about 80 languages are spoken.
- According to the 2007 Population and Housing Census of Ethiopia, Afan Oromo (33.8%) and Amharic (29.3%) were the major mother tongues in the country
- Somaligna (6.2%), Tigrigna (5.9%), Sidamigna (4.0%), Wolaytigna (2.2%), Guragigna (2%), Afarigna (1.7%), Hadyiyagna (1.7%), and Gamogna (1.5%) do have significant number of speakers.
- On the other hand, some of the languages are spoken by a few thousands

- The Ethiopian languages belong to two Supper Families: Afro-Asiatic and Nilo-Saharan. Most Ethiopian languages belong to the Afro-Asiatic Supper Family.

1. Afro-Asiatic

- The Afro-Asiatic Supper Family, is divided into three families, namely: **Semitic**, **Cushitic** and **Omotic**
- The **Cushitic** languages are predominantly spoken in central, southern, eastern and northeastern parts of Ethiopia mainly in Afar, Oromia and Somali Regional States. It has the largest number of speakers and the widest spatial coverage. It includes Oromigna, Somaligna, Sidamigna, Afarigna, Kembatigna, Hadiyigna, Alabigna, Gedeogna, and others.
- The **Semitic** languages are spoken in northern, central and eastern parts of Ethiopia particularly in the regional states of Tigray, Amhara, Harari and northern SNNPR. Some of the Semitic Languages include Amarigna, Tigrigna, Guragigna, Siltigna, Aderigna, and Argobigna.

- The **Omotic languages** are predominantly spoken in the south –central and south-western parts of Ethiopia mainly between the Lakes of southern Rift Valley and the Omo River. The languages, which make up this family, are numerous which include Wolaitigna, Gamogna, Kullogna, Kefigna, and Kontigna etc.

2. NiloSaharan

- The Nilo-Saharan languages are spoken in the western lowlands of Ethiopia along the border with Sudan, in Gambella and Benishangul Gumuz Regional States. They include Kunamigna, Bejigna, Gumuzigna, Maogna, Kewamigna, Nuerigna, Annukigna, and others.

Settlement Types and Patterns

Types of Settlement

- Settlements are places that are inhabited by people more or less on a permanent basis, as distinct for example from camps, and where people carry out a variety of activities such as agriculture, manufacturing and commerce
- Settlements are divided into two, namely, rural and urban on the bases of the dominant economic activity, population densities and availability of socioeconomic and infrastructural facilities.

Rural Settlement

- The vast majority of the Ethiopian population still lives in rural settlements consisting of hamlets and villages.

1. Temporary / Mobile Settlements

- The lowlands in most parts of the Rift Valley and peripheral areas, being generally hot and dry, are characterized by pastoral herding and mobile settlements.
- The major problem often mentioned about mobile settlements is that of providing social services like clean water, schools, hospitals, electricity etc to the people.

• 2. Permanent Settlements

- Settlements are considered as permanent if there are no frequent changes in their locations. Most rural highland settlements where crop cultivation is practiced are permanent.
- Permanent settlements could be **scattered**/diffused/dispersed), or **grouped**/ clustered/ nucleated

Urban Settlements and Urbanization in Ethiopia

- **Urbanization:**
 - refers to the increase in the percentage of the population living in urban centers.
 - the process of becoming urban, moving to cities and changing from agriculture to other pursuits of life which are common to towns and, with a corresponding change of behavioral patterns.
 - is crucial to sustain the pace of economic development and improve the quality of life for both urban and rural populations.
- Linkage between urban and rural areas could foster efficiency of value chains in agro-industry, improve agricultural productivity, promote service expansion and create sufficient industrial jobs in urban centers to absorb the perpetual influx of population from rural areas.

However, mismanaged rapid urbanization could result in:

- unemployment,
- housing shortages and informal settlements,
- infrastructural and service shortages,
- poverty and social distress.

The major criteria to classify settlements as urban in Ethiopia are:

- ❖ Minimum of 2,000 people;
 - ❖ Two-thirds of the population engaged in non-agricultural activities;
 - ❖ Chartered municipality;
 - ❖ The presence of social services and amenities
- The number of settlements meeting these criteria in 1984 was 322; comprising 10.23% of the total population of the country
 - The number of settlements with greater than 2,000 people in 1994 had increased to 539; consisting 12.8% of the country's population.
 - In 2007, the number further rose to 927. Today, the urban population is about 20 percent of the country's population.

An overview of the History of Urbanization in Ethiopia

- Before the foundation of Addis Ababa as a capital city, the earliest capitals and other towns did not have a permanent population exceeding 6000; had little impact on the hinterland
- The only prominent urban centers were Axum, Lalibela and Gondar.
- There were wandering capitals and mobile military camps for long that followed their peripatetic rulers.
- Modern urbanization in Ethiopia is associated with the establishment of **Addis Ababa as a capital by Emperor Menelik II in the late 19th century.**

Conditions contributing to growth & permanency of Addis Ababa as a capital:

- Introduction of the fast-growing Australian eucalyptus tree which solved the firewood needs of the ever-growing urban population.
- Water supplies improved due to introduction of wells & reservoirs.
- Introduction of modern schools, hospitals and health centres, hotels, cafes, bars, bakeries, butcher's shops, cinema, post office, modern cathedrals, telephone and telegraph system, bank, printing press etc.
- The construction of roads that radiate from Addis Ababa; and the arrival of the Franco-Ethiopian railway at Addis Ababa in 1917.
- The Italian occupation had intensified the establishment of small-scale industries and institutions, road construction

- Today, Addis Ababa is the capital of Ethiopia; diplomatic capital of Africa; seat of numerous embassies and consular representatives; and UN ECA headquarters.
- Addis Ababa is geographically at the very heart of Ethiopia.
- It has a congenial climatic condition; and enjoys excellent connections with all Ethiopia's economic zones. It is also connected to the rest of the world via its reputed airlines and various international airlines.

Ethiopia is a least urbanized country: 'vast urban desert'

Ethiopia's low level of urbanization can be attributed to:

- The self-sufficiency of agriculture which reinforced rural peasant life;
- Low level of industrialization, low level structural transformation and economic development
- The morphology of the country that hindered transportation and communication
- The continual warfare for centuries between kingdoms; and the frequent changes of the royal residence
- Lack of employment, and housing shortage in urban areas that discourage in migration;
- Political instability, ethnic conflict and social unrest during the late 1970s and 1980s;

Drivers of and Opportunities for more Urbanization in Ethiopia

- The urban population is growing at 5% a year, and the 20% urban pop. is projected to mount to 37% by 2035. Factors:
 - Establishment of **Addis Ababa** as centre of expansion & its permanency.
 - The construction of the **Ethio-Djibouti railway** .
 - The **five-year Italian occupation** .
 - **All-weather roads** radiating from Addis .
 - **Political decentralization** & provision of administrative status
 - **Proximity to existing cities** & main transportation corridors
 - **High Population density** and growth rates in the populous highlands
 - New and large **commercial farms, mining areas, and agro-industries**
 - **Large infrastructure investments** such as airports, highways & dry ports
 - Opening of **Universities** in some localities
 - Tourism assets & attractions such as parks, resort centres, and heritage cities
 - Development of border towns with strengthened inter-country trade

Distribution of Urban Centers in Ethiopia

- Urban centers in Ethiopia are spatially unevenly distributed owing to varying concentration of industries; and services such as schools, health institutions, water supplies, electricity, means of transport, etc.

1. *The Addis Ababa Metropolitan cluster* includes Addis Ababa and its surrounding towns; & Adama and its surrounding towns

2. *Secondary city clusters:*

- Lake Tana Urban Cluster: Bahir Dar, Gondar Debre Tabour, Debre Markos
- South Rift Valley Urban Cluster: Hawassa-Shashemene-Dila; and Hosana-Sodo-Arba Minch
- Eastern Urban Cluster: Dire Dawa, Harar, Jigjiga
- Mekelle Urban Cluster: Mekelle, Adigrat, Shire, Axum
- Dessie- Kombolcha Urban Cluster.
- Jima Urban Cluster: Jima, Agaro, Mizan, Tepi, Gambella

3. *Tertiary urban clusters* :

- Nekemte Urban Cluster: Nekemte, Dembidolo, Gimbi, Metu, Assosa
 - Gode - Kebri Dar oasis city network
 - Semera-Mille – Asaita oasis city network
- ❑ Like most developing countries, Ethiopia's urban population is concentrated in one primate city, Addis Ababa (about 29%)

Growth Rate of Urban Centers

- Ethiopia's towns are characterized by spatiotemporal growth rates (5% annual average) variations.
- 1. **Declining Towns**: includes towns whose populations are declining in absolute numbers because net out migration is greater than natural increase.
 - ❑ Some of the towns that have at least once experienced a decline include, Axum, Goba and Maichew.
- 2. **Slow Growing Towns**: includes towns that grow at the rate which is less than the rate of natural increase.
 - ❑ Towns such as Holeta, Harar and Gore have been indicated to grow slowly in the recent past.
- **Fast Growing Towns**: Includes all towns with growth rates of greater than the natural rate of increase; owing to large in migration. Examples include Addis Ababa, Hawassa, Bahir Dar....

Thank you!